ECE 647: Lecture 5

Alternate proof: by definition.

3. Convexity of the Set

The set
$$S = \left\{ X \in \mathbb{R}^m \;\middle|\; |P_X(t)| \le 1 \text{ for all } |t| \le \frac{\pi}{d} \right\}$$
where
$$P_X(t) = \sum_{k=1}^{m/2} X_k \cos kt$$
is convex.

Proof

For a given t,
$$\{X \mid |P_X(t)| \le 1\}$$
is equivalent to
$$\left\{ X \;\middle|\; -1 \le \sum_{k=1}^{m/2} X_k \cos kt \le 1 \right\}.$$
Since this is the intersection of two half-spaces, it follows that S is convex.

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4. Union of Convex Sets

Q: Is the union of convex sets a convex set?

A: Usually not.

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is affine if it is of the form

$$f(x) = Ax + b, \quad A \in \mathbb{R}^{m \times n},\ b \in \mathbb{R}^m.$$

If $S \subseteq \mathbb{R}^n$ is convex, and $f : \mathbb{R}^n \to \mathbb{R}^m$ is affine, then the image of S under f,

$$f(S) = \{f(x) \mid x \in S\}$$

is convex.

Intuition

- Straight lines map to straight lines.
- If x_1 can reach x_2 via straight lines without leaving S, then $f(x_1)$ must reach $f(x_2)$ via straight lines without leaving $f(S)$.

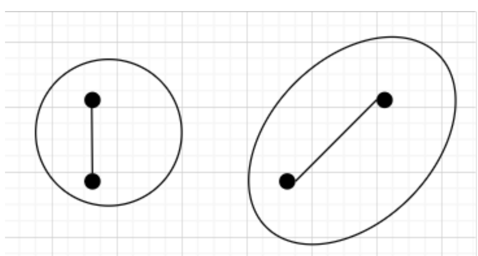


Figure 1: Affine transformation of a ball into an ellipsoid.

Examples

- Balls and ellipsoids.

A ball is defined as:

$$B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\}$$

where $\|\cdot\|_2$ denotes the Euclidean norm.

An ellipsoid (a more restrictive version) is defined as:

$$\mathcal{E} = \{x \mid (x - x_c)^T P(x - x_c) \leq 1\}$$

where P is a positive definite matrix.

Then, $\mathcal{E}$ is the image of the unit ball $B(0, 1)$ under the mapping:

$$f(u) = P^{1/2}u + x_c.$$

Why?

$$(f(u) - x_c)^T P^{-1}(f(u) - x_c) = u^T P^{1/2} P^{-1} P^{1/2} u = u^T u \leq 1$$

if $u \in B(0, 1)$.

Thus, $\mathcal{E}$ is convex.

The third kind of operations that preserve convexity is the *perspective function* & *linear-fractionals*.

Write

$$X = (z, t), \quad \text{where } z \in \mathbb{R}^n, \text{ and } t \in \mathbb{R}^{++}$$

(positive real numbers).

Let

$$p(x) \triangleq p(z, t) = \frac{z}{t} \in \mathbb{R}^n.$$

Then $p(.)$ is called a *perspective function*.

Intuition

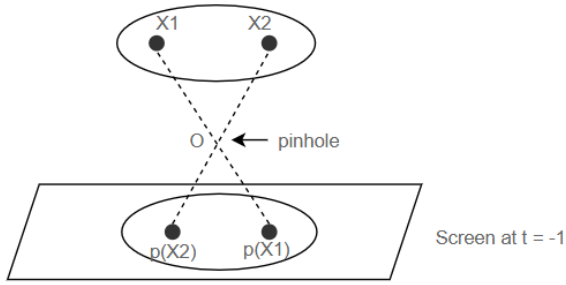
- Scale the vector such that the last component is 1.
- Then drop the last component.
- Can be interpreted as the action of a *pin-hole camera*.

If $C \subset \mathbb{R}^n \times (\mathbb{R}^+ \setminus \{0\})$ is convex, then

$$p(C) = \{p(x) \mid X \in C\}$$

is also convex.

Proof: See Boyd P40. Again, straight lines are mapped to straight lines.



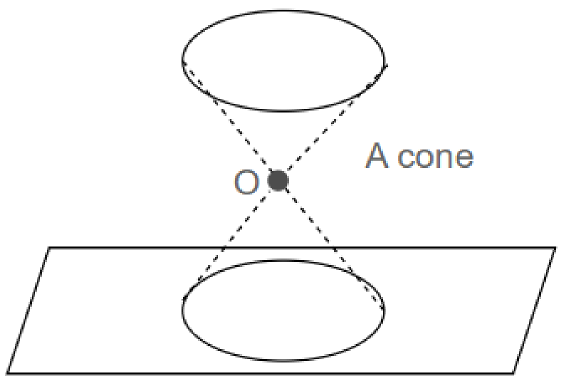
The Inverse is also true

If C is convex, define

$$p^{-1}(C) = \left\{(x,t) \in \mathbb{R}^{n+1} \mid \frac{x}{t} \in C, \quad t > 0\right\}.$$

Then $p^{-1}(C)$ is convex.

Proof: See Boyd P40.



Linear Functional

$$f(x) = \frac{Ax+b}{Cx+d}$$

where $A \in \mathbb{R}^{m\times n}$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^n$, and $d \in \mathbb{R}$.

If

$$X = \{x \mid Cx + d > 0\}$$

is convex, then $f(X)$ is convex.

Proof

Write $f(x)$ as:

$$f(x) = P(g(x))$$

where

$$g(x) = (Ax + b, Cx + d)$$

is an affine transformation, and $P(w)$ is the *perspective mapping*.

Convex Sets

A set C is convex if for any $x_1, x_2 \in C$ and any θ such that $0 \leq \theta \leq 1$, we have:

$$\theta x_1 + (1 - \theta) x_2 \in C$$

Key Examples

- Affine sets, subspaces
- Balls, ellipsoids
- Hyperplanes, half-spaces
- Polyhedra

Operations that Preserve Convexity

- Intersections
- Affine mappings
- Perspectives